

DSAA3073: Theories in Data Science

Week 8 Validation Sheet: Online Learning: Regret Analysis and Algorithms

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Question 1. [5 marks] In the online learning framework, the learner's goal is to:

- a) Minimize training error on a fixed dataset
- b) Minimize regret compared to the best fixed strategy in hindsight
- c) Learn a hypothesis that generalizes to unseen test data
- d) Maximize the likelihood of the observed data

Question 2. [5 marks] The Halving Algorithm eliminates experts that:

- a) Have the lowest weight
- b) Make a mistake in the current round
- c) Have made more than one mistake total
- d) Disagree with the majority prediction

Question 3. [5 marks] For Weighted Majority with $\beta = \frac{1}{3}$, if an expert has weight $w = 9$ and makes two consecutive mistakes, its new weight becomes:

- a) 3
- b) 1
- c) 6
- d) 0

Question 4. [5 marks] Which statement about regret bounds is TRUE?

- a) $O(\log T)$ regret means the algorithm performs worse than the best expert
- b) $O(\sqrt{T})$ regret means average per-round regret grows without bound
- c) $O(T)$ regret means the algorithm's average per-round performance approaches the best expert's
- d) $O(\sqrt{T})$ regret means average per-round regret goes to zero as $T \rightarrow \infty$

Question 5. [5 marks] The optimal learning rate for Randomized Weighted Majority when T is known satisfies:

- a) Larger η is always better for faster learning
- b) η should increase with T to ensure convergence
- c) η should decrease with T to balance exploration and exploitation
- d) η is independent of the number of experts n

Question 6. [5 marks] In MWU with losses $m_t(i) \in [0, 1]$, what does $\langle p_t, m_t \rangle$ represent?

- a) The loss of the best expert at round t
- b) The maximum loss among all experts at round t
- c) The expected loss of the algorithm at round t
- d) The cumulative regret up to round t

Question 7. [5 marks] The key advantage of randomized algorithms over deterministic ones in adversarial online learning is:

- a) Randomized algorithms are computationally faster
- b) The adversary cannot predict the algorithm's exact action before committing to outcomes
- c) Randomized algorithms require less memory
- d) Randomized algorithms always outperform all experts

Question 8. [5 marks] The regret bound for MWU includes a $\ln n$ term because:

- a) The algorithm needs $\ln n$ rounds to identify the best expert
- b) The potential function analysis uses the ratio $\frac{\Phi_1}{\Phi_{T+1}} \geq \frac{1}{n}$
- c) There are $\ln n$ possible weight values
- d) The learning rate must be at least $\frac{1}{\ln n}$

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Consider $n = 4$ experts over $T = 5$ rounds with binary predictions:

Round	Expert 1	Expert 2	Expert 3	Expert 4	Outcome
1	1	0	1	0	1
2	0	1	1	1	0
3	1	1	0	1	1
4	0	0	0	1	0
5	1	0	1	0	1

- (a) [6 marks] For each expert, count the total number of mistakes. Which expert is best?
- (b) [4 marks] If we always followed the best expert from the start, how many mistakes would we make?
- (c) [5 marks] Compute the regret bound for Weighted Majority with $\beta = \frac{1}{2}$. Would the algorithm outperform always following a single random expert?

Question 10. [13 marks] The Randomized Weighted Majority algorithm samples expert i with probability proportional to its weight.

(a) [7 marks] Suppose at round t we have three experts with weights $w_t(1) = 8$, $w_t(2) = 4$, $w_t(3) = 2$. Compute the sampling probability for each expert. What is the expected number of mistakes if their losses are $m_t(1) = 0$, $m_t(2) = 1$, $m_t(3) = 0$?

(b) [6 marks] After observing these losses, compute the new weights $w_{t+1}(i)$ using learning rate $\eta = 0.25$. What are the new sampling probabilities?

Question 11. [15 marks] Prove the lower bound for deterministic online learning algorithms.

- (a) [8 marks] Show that for any deterministic algorithm $\mathcal{A}$ in the binary prediction problem, there exists an adversarial strategy that forces $\mathcal{A}$ to make at least $\frac{T}{2}$ mistakes over T rounds.
- (b) [7 marks] Explain why this lower bound does not apply to randomized algorithms. What property of randomization breaks the adversarial strategy?

Question 12. [12 marks] Consider the potential function $\Phi_t = \sum_{i=1}^n w_t(i)$ in the MWU analysis.

(a) [6 marks] Show that $\Phi_{T+1} \geq w_{T+1}(i^*)$ for any expert i^* . Use this to derive a lower bound on Φ_{T+1} in terms of the best expert's cumulative loss.

(b) [6 marks] Combining the upper bound $\Phi_{T+1} \leq n \exp\left(-\eta \sum_{t=1}^T \langle p_t, m_t \rangle\right)$ with your lower bound, derive the MWU regret bound.

Question 13. [10 marks] (a) [5 marks] Explain why the regret bound has a $\sqrt{T}$ dependence. Why can't we achieve $O(\log T)$ regret in the adversarial setting?

(b) [5 marks] In what scenarios might we achieve $O(\log T)$ regret? What additional assumptions are needed?

Question 14. [15 marks] Compare and contrast online learning with PAC learning across multiple dimensions.

- (a) [8 marks] For each framework, describe: (i) the data assumption, (ii) the performance measure, (iii) the type of guarantees provided, and (iv) a typical application domain.
- (b) [7 marks] Despite their differences, both frameworks achieve sublinear scaling in some parameter. Explain what this means in each case and why it indicates successful learning.

